

BAAZ JKBOSE

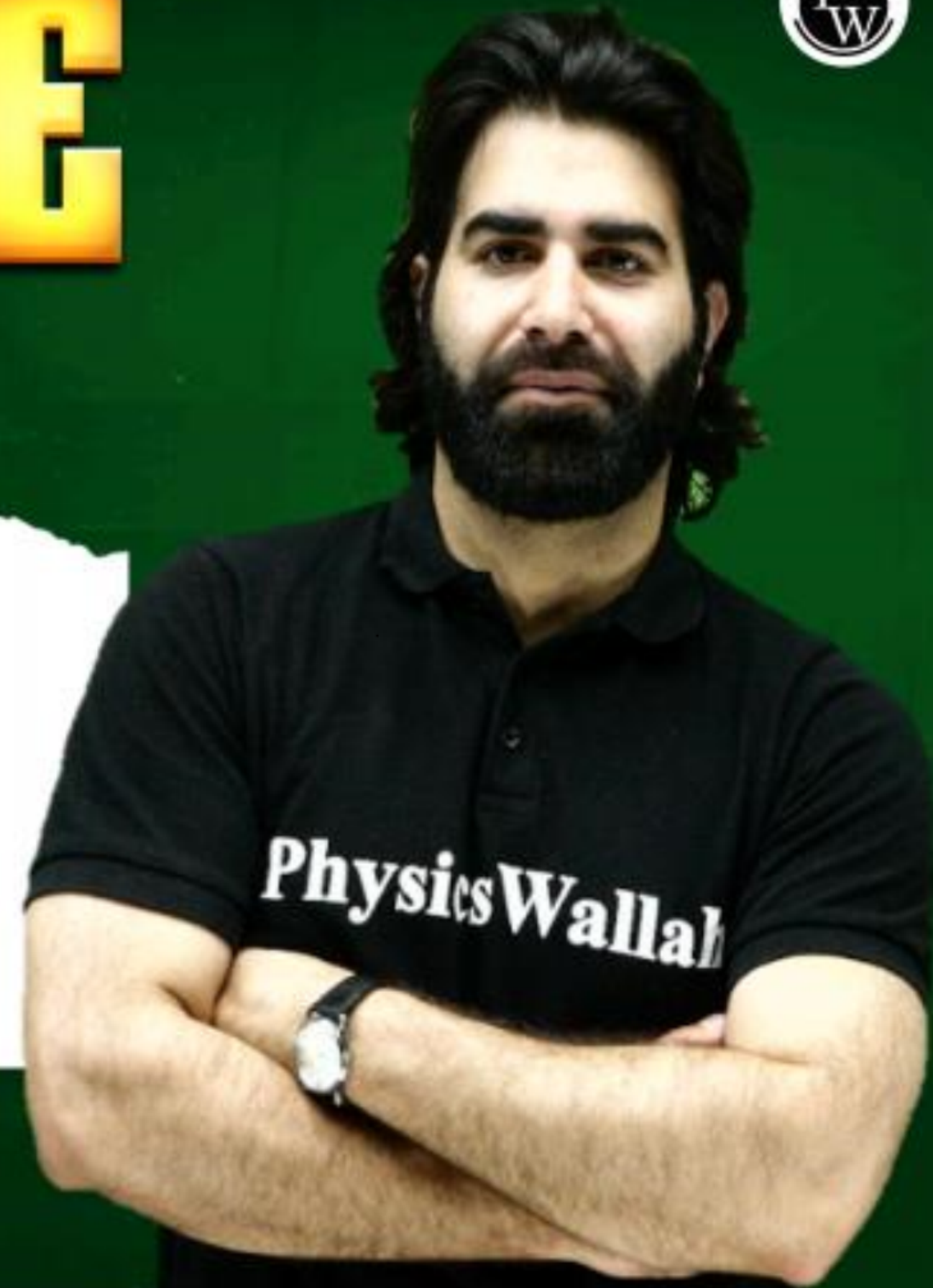
2026 CLASS-11TH

Locomotion and
Movement

LECTURE-4

Zoology

Dr Adil Altaf sir



Topics To Be Covered

- 1 Skeletal System

Myoglobin - Oxygen Storing Protein (Hold)

T/W



Red Muscle Fibre

Reddish
Appearance

→ Rich in Myoglobin (Stores O_2)
→ More Blood Supply (More O_2)

→ Show Aerobic Respiration
often, Known as Aerobic
Muscle.

(Also shows fatigue).

→ Slow contraction.
[slow twitch].

→ Sarcoplasmic Reticulum less
developed

⊙ Muscle of Back, limbs etc

White Muscle Fibre

→ Less in Myoglobin.

→ Less Blood Supply.

→ Fatigue easily, as they go
into Anaerobic Respiration easily
- klas - Anaerobic Muscle.

Fast contraction (Fast twitch Muscle)
Fibre)

→ S.R. well developed.

⊙ Muscles of Eye ball.

Source of Energy in Muscles

① Stored ATP: $(\text{ATP} \rightarrow \text{ADP} + \text{P})$

② Creatine Phosphate - $\text{CP} + \text{ADP} \longrightarrow \text{ATP} + \text{Creatine}$

[creatine metabolises gradually - creatinine (waste)]

③ New ATP

Two ways

Continuously

Aerobic

- O₂ Present

Steps → Glycolysis

→ Krebs Cycle

→ Net ATP = 36/38

Depends

- Type of Muscle

- Load

- Duration

Anaerobic

- O₂ Absent

Steps - Glycolysis → 8ATP

- Lactic Acid Fermentation

Glucose

↓
Pyruvic Acid

Pyruvic Acid →

Lactic Acid

↓
H⁺

Accumulation → Fatigue

Blood → Liver

Coxis Cycle

Net ATP = 2ATP

~ Muscle Fatigue -

- Decreased efficiency of Muscle to generate power for contraction.
- Due to Accumulation of products of Anaerobic Respiration like lactic Acid, H^+ , etc

- Cori's Cycle - Conversion of Lactic Acid Back into Glucose (Glycogen)
occurs Liver.

Human Skeletal System

↳ Consists of Bone & Cartilage

↳ It is for, Posture, protecting organs, Movement, Blood formation etc

Bone

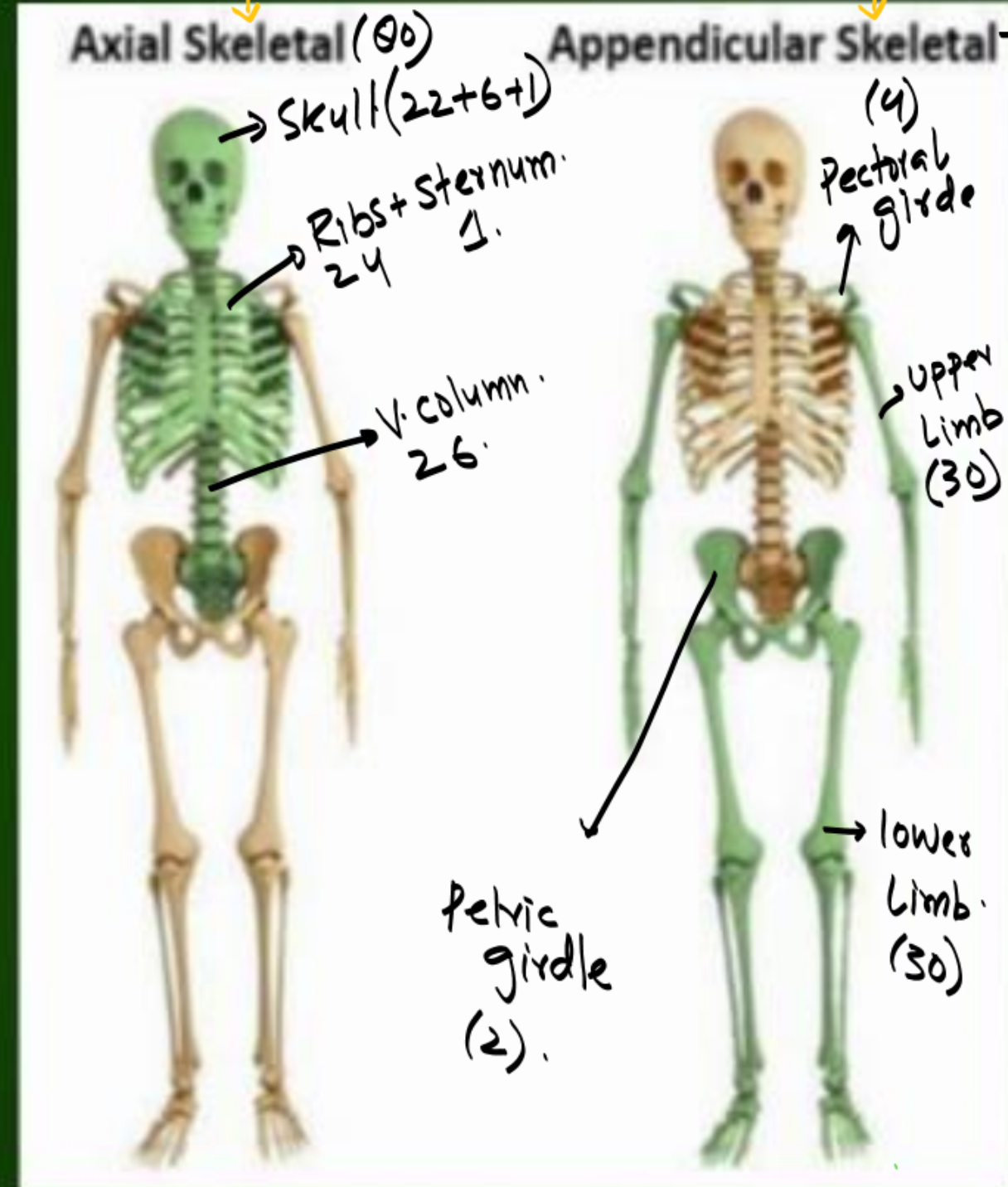
- Hard, Non-Pliable
- Matrix - Calcium, phosphate Salts
- Cell - Osteocyte, Osteoblast (Young)

Cartilage

- Soft, Pliable (Flexible)
- Matrix - without Salts (except - Calcified Cartilage)
- Chondroblast (Young)
- Chondrocyte

Adult Human Skeleton

(206 Bones)



Thank
You